

Name: _____

Date: _____

PERCENTAGES OF AMOUNTS, (FINDING ANY PERCENTAGE OF AN AMOUNT)

LO: I can find any percentage of an amount using appropriate methods.

Methods for finding percentages

To find **any percentage** of an amount, you can use the **decimal method** (convert % to a decimal and multiply) or the **building-up method** (use 10%, 5%, 1% blocks).

Calculator method: amount $\times$ percentage $\div$ 100. Or simply amount $\times$ decimal.

Quick examples

●	23% of 400 = $400 \times 0.23 = 92$	decimal method
●	17% of 300 = $30 + 21 = 51$	10% + 7%
●	8.5% of 200 = 17	200×0.085
●	23% of 400 = $23 \times 4 = 92$ but method was % $\div$ amount	method matters
●	0.7% of 1000 = 7	0.7% = 0.007, answer is 7

Key idea

Convert the percentage to a decimal by dividing by 100, then multiply by the amount.

$$\text{23\% of 400} = 0.23 \times 400 = \text{92}$$

Percentage $\div$ 100 = decimal multiplier

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - decimal method

Find 34% of £650.

$$\begin{aligned} \text{Convert: } 34\% &= 0.34 \\ 650 \times 0.34 \\ &= \text{£221} \end{aligned}$$

£221

Convert the percentage to a decimal and multiply.

Example 2 - building-up method

Find 17% of £400.

$$\begin{aligned} 10\% \text{ of } 400 &= 40 \\ 5\% &= 20; 2\% = 8 \\ 17\% &= 40 + 20 + 8 - 12\% \text{... Actually:} \\ 10\% + 5\% + 2\% &= 40 + 20 + 8 = 68 \end{aligned}$$

£68

Build percentages from 10%, 5%, and 1% blocks.

Example 3 - small percentage

Find 3.5% of £2000.

$$\begin{aligned} 1\% \text{ of } 2000 &= 20 \\ 3\% &= 60; 0.5\% = 10 \\ 3.5\% &= 60 + 10 = \text{£70} \end{aligned}$$

£70

For decimal percentages, use 1% and halves as building blocks.

Example 4 - in context

A bank charges 2.8% interest on a £5000 loan. Find the annual interest.

$$\begin{aligned} \text{Convert: } 2.8\% &= 0.028 \\ 5000 \times 0.028 \\ &= \text{£140} \end{aligned}$$

£140

Always multiply by the decimal equivalent of the percentage.

YOUR TURN – PRACTICE (deliberate practice)

1. Decimal method

Use the decimal method to find each percentage.

- a) 14% of £300
- b) 23% of £500
- c) 37% of £200
- d) 9% of £450

2. Building-up method

Use 10%, 5% and 1% to calculate.

- a) 16% of £400
- b) 21% of £300
- c) 35% of £180
- d) 12% of £550

3. Decimal percentages

Find each percentage.

- a) 2.5% of £800
- b) 7.5% of £240
- c) 0.5% of £6000
- d) 4.5% of £1000

4. Larger amounts

Calculate each percentage.

- a) 18% of £3500
- b) 6% of £12,000
- c) 22% of £4500
- d) 3% of £25,000

5. Real-world problems

Solve each problem.

- a) A £680 TV has 12% off. Find the discount.
- b) Commission is 3.5% of £8000 sales. Find the commission.
- c) Tax is 19% on £42,000 income. Find the tax.
- d) A 7% tip on a £55 meal.
- e) Insurance is 4.2% of a £15,000 car value.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To find 23% of a number, multiply by 0.23 ☐

Correct answer: _____

Reason: _____

b) 0.5% of 1000 = 50 ☐

Correct answer: _____

Reason: _____

c) 17% can be found as 10% + 5% + 2% ☐

Correct answer: _____

Reason: _____

d) 8% as a decimal is 0.8 ☐

Correct answer: _____

Reason: _____

e) Finding 1% means dividing by 100 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A company makes **£84,000** profit. They pay **19% corporation tax**, then distribute **35%** of the remaining profit as dividends. How much is paid in dividends?

Expression: _____

Simplified: _____

b) Three shops sell the same TV. Shop A: **£600 with 15% off**. Shop B: **£550 with 8% off**. Shop C: **£520 with 2% off**. Which shop offers the best deal?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

